```
Rules for Identifiers
Identifiers can be combination of uppercase and lowercase letters, digits or an underscore (_).
So myVariable, variable_1, variable_for_print all are valid python identifiers.
An Identifier can not start with digit.
So while variable1 is valid, 1variable is not valid.
We can’t use special symbols like !, #, @, %, $ etc in our identifier.
Identifier can be of any length.
```

In [116]:
```
#Variable

name="sheela"
age=25
address="USA"

print(f"I am {name}, i am {age} and i am from {address}")
```

```
I am sheela, i am 25 and i am from USA
```

In [117]:
```
name="sheela"
age=25
address="USA"

print(name)
```

```
sheela
```

You don’t need to tell Python the data type. It figures it out on its own.

In [118]:
```
name="sheela"
age=25
address="USA"

print(type(name))
print(type(age))
print(type(address))
```

```
<class 'str'>
<class 'int'>
<class 'str'>
```

Data type

Integers string float Boolean T/F

Cannot be used as variable names (keywords)

These are reserved in Python:

if else elif for while break continue def return True False None and or not in is import from as try except finally with pass

In [119]:
```
a=5
b=6
sum=a+b
print(sum)
```

```
11
```

In [120]:
```
a=1000
b=500
diff=a-b
print(diff)
```

```
500
```

In [121]:
```
#type conversion -conversing and casting
```

In [122]:
```
a=6   #since float is superior and provide extra data , it converts int into float.
b=8.5
sum= a+b
print(sum)
```

```
14.5
```

In [123]:
```
a=int("2") #type casting . #it wont work for all data , only those which if the another data can be fit it
b=4.65
sum=a+b
print(sum)
```

```
6.65
```

Input in Python

input() statement is used to accept values (using keyboard) from user.

input() → result for input() is always a string (str)

int(input()) → used to convert input into integer

float(input()) → used to convert input into float

In [124]:
```
input("enter your name:")
```

```
enter your name:sheela
```

Out[124]:
```
'sheela'
```

In [125]:
```
name=input("enter your name:")
print("Welcome",name)
```

```
enter your name:sheela
Welcome sheela
```

In [126]:
```
int("5")
val=float(input("enter some value;"))
print(type(val),val)
```

```
enter some value;7
<class 'float'> 7.0
```

In [127]:
```
name = input("enter your name:")
age=int(input("enter your age:"))
mark=float(input("enter mark"))
print("welcom",name)
print("age",age)
print("mark",mark)
```

```
enter your name:sheela
enter your age:25
enter mark70
welcom sheela
age 25
mark 70.0
```

In [128]:
```
a=int(input("Enter the number1:"))
b=int(input("Enter the number2:"))
print("sum:",a+b)
```

```
Enter the number1:5
Enter the number2:7
sum: 12
```

In [129]:
```
side=float(input("enter squar side:"))
print("area=", side**2)
```

```
enter squar side:6
area= 36.0
```

Strings

String is a data type that stores a sequence of characters.

Basic Operations

concatenation "hello" + "world" → "helloworld"

length of string len(str)

In [130]:
```
str1="this is a string"
str2='python'
str3="""this is string"""
print(str1)
```

```
this is a string
```

In [131]:
```
str1="This is a string.\nwe are creating it in python." #\n #\t
print(str1)
```

```
This is a string.
we are creating it in python.
```

In [132]:
```
#CONCATENATION
str1="sheela"
str2="kunwar"
final_str=str1+" "+str2
print(final_str)
len(final_str)
```

```
sheela kunwar
```

Out[132]:
```
13
```

In [133]:
```
str="sheela kunwar"
ch=str[0]
ch1=str[3]
print(ch)
print(ch1)
```

```
s
e
```

string- Slicing Accessing parts of a string

str[starting_idx : ending_idx] → ending index is not included

str = "Sheela Kunwar"

str[1 : 4] is "hee"

str[:4] is same as str[0 : 4] → "Shee"

str[1 :] is same as str[1 : len(str)] → "heela Kunwar"

In [134]:
```
str= "sheela kunwar"
print(str[1:6])
```

```
heela
```

In [135]:
```
str= "sheela kunwar"
print(str[5:])
```

```
a kunwar
```

In [136]:
```
str= "sheela kunwar"
print(str[:4])
```

```
shee
```

In [137]:
```
#negative index
```

In [138]:
```
str="apple"
print(str[:-2])
```

```
app
```

String Functions

str = "I am a coder."

str.endswith("er.") → returns True if string ends with er

str.capitalize() → capitalizes first character

str.replace(old, new) → replaces all occurrences of old with new

str.find(word) → returns index of first occurrence

str.count("am") → counts occurrence of substring in string

In [139]:
```
str="I am studing python from youtube"
print(str.endswith("tube"))
```

```
True
```

In [140]:
```
str="i am studing python from youtube"
print(str.capitalize())
print(str)#doesnt change old string
```

```
I am studing python from youtube
i am studing python from youtube
```

In [141]:
```
str="i am studing python from youtube"
str=str.capitalize()
print(str)
```

```
I am studing python from youtube
```

In [142]:
```
str1="i am studing python from youtube"
print(str1.replace("python","java"))
```

```
i am studing java from youtube
```

In [143]:
```
str="my name is sheela"
print(str.replace("sheela","seema"))
```

```
my name is seema
```

In [144]:
```
#FIND
str1="i am studing python from youtube"
print(str1.find("python"))
```

```
13
```

In [145]:
```
#count
str1="i am from studing python from youtube"
print(str1.count("from"))
```

```
2
```

In [146]:
```
# WPA to input user's first name and print its length
name=input("enter your name:")
print("length of your name:",len(name))
```

```
enter your name:sheela kunwr
length of your name: 12
```

In [147]:
```
#WPA to find the occurance of $ in the string.
a="a","e","d","$","$"
print(a.count("$"))
```

```
2
```

Conditional Statement if-elif-else

syntax if(condition): statement1 elif(condition): statement2 else: statementN

In [148]:
```
age=int(input("enter the age:"))

if age>=18:
    print("can vote and apply for license")
else:
    print("you can't vote and apply for license")
```

```
enter the age:25
can vote and apply for license
```

In [149]:
```
light = "green"

if(light=="red"):
    print("stop")
elif(light=="yellow"):
    print("look")
elif(light=="green"):
    print("go")
else:
    print("light is broken")
```

```
go
```

In [150]:
```
mark=int(input("Enter your mark"))

if mark >=90:
    print("grade A")
elif mark>=80:
    print("grade B")
elif mark >=70:
    print("grade C")
else:
    print("grade D")
```

```
Enter your mark67
grade D
```

In [151]:
```
#nesting - condition inside another condition
age=int(input("enter the age:"))

if age>=18:
    if (age>=80):
        print("cannot drive")
    else:
        print("can drive")
else:
    print("cannot drive")
```

```
enter the age:55
can drive
```

In [152]:
```
num=14
rem= num%2

if (rem==0):
    print("even")
else:
    print("odd")
```

```
even
```

In [153]:
```
a = int(input("Enter the number"))
b= int(input("Enter the number"))
c = int(input("Enter the number"))

if (a>=b and a>=c):
    print(" a is greater",a)
elif (b>=c):
    print("b is greater", b)
else:
    print("c is greater" ,c)
```

```
Enter the number6
Enter the number7
Enter the number9
c is greater 9
```

list slicing

In [154]:
```
marks=[22,33,44,55,66,77,88]
print(marks[:4])
```

```
[22, 33, 44, 55]
```

In [155]:
```
marks=[22,33,44,55,66,77,88]
print(marks[-1:])
```

```
[88]
```

```
tuple = a built-in data that lets us creatw immutable sequencw of values.
```

In [106]:
```
tup=(2,3,4,5,6)  #doesnt support the value assignment.
print(type(tup))
```

```
<class 'tuple'>
```

```
#tuple methods
tup=(2,1,3,1)
tup.index(el)#returns index of first occurrence tup.index(1) is 1
tup.count(el)#counts total occurances tup.count(1) is 2
```

In [110]:
```
tup=(2,3,4,5,6)
print(tup.index(4))
```

```
2
```

In [111]:
```
tup=(2,3,4,5,6,6)
print(tup.count(6))
```

```
2
```

In [114]:
```
flowers= []
flower1= input("Enter your favorite flower")
flower2= input("Enter your favorite flower")
flower3= input("Enter your favorite flower")
flowers.append(flower1)
flowers.append(flower2)
flowers.append(flower3)

print(flowers)
```

```
Enter your favorite flowerrose
Enter your favorite flowersunflower
Enter your favorite flowerlilly
['rose', 'sunflower', 'lilly']
```

In [115]:
```
movies=[]
movies.append(input("enter the 1st flower name"))
movies.append(input("enter the 1st flower name"))
movies.append(input("enter the 1st flower name"))
print(movies)
```

```
enter the 1st flower namelilly
enter the 1st flower namerose
enter the 1st flower namelavender
['lilly', 'rose', 'lavender']
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: